Jonathan Poteet

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PROFESSIONAL SUMMARY

Software Engineer with 5+ years of experience shipping production-grade applications. Expert in TypeScript, Python, and Java with a specialized focus on Frontend Architecture, CI/CD Automation, Formal Verification, and high-performance distributed systems. Lead technical contributor for GM's enterprise web platform modernization, expert in scaling micro-frontends and ensuring system reliability through formal methods and security-first development.

CORE COMPETENCIES

Frontend Architecture | Micro-frontends (Angular/Solid/React) | Design-to-Code Fidelity | CI/CD & DevOps Automation | Formal Verification (PRISM/StormPy) | Application Security (Security+) | Cloud Orchestration (Kubernetes) | Responsible AI & ML Pipelines | Procedural Generation

WORK EXPERIENCE

General Motors
March 2021 - December 2025

Software Engineer - Roswell, GA

- **Software Systems Modernization:** Lead technical contributor for the full-stack modernization of enterprise technician and customer web platforms (Service Workbench, D2C). Re-engineered legacy architectures using Angular, TypeScript, Next.js, and Java, improving scalability and responsiveness for a global user base.
- **Systems Architecture & Containerization:** Architected Docker and Podman container environments using WSL to build and manage complex microservices in multi-repository Linux environments, achieving 100% environment parity and reducing deployment errors.
- **Project Leadership & International Deployment:** Directed frontend development of the "My Orders" platform for four European markets. Integrated Strapi CMS and enterprise APIs to deliver real-time, localized vehicle/financial data while ensuring international compliance.
- **DevOps & Systems Interoperability:** Optimized API discoverability using Spring Boot Swagger. Managed end-to-end CI/CD pipelines via Azure DevOps, automating software delivery and reducing manual deployment time.
- **Quality Assurance & Performance:** Led the migration of the MyOrders platform to Solid.js within the DASH monorepo. Established rigorous code quality standards, including Angular and ESLint Strict Modes, ensuring long-term stability and security.
- **Technical Innovation:** Integrated Vitest for comprehensive automated testing and utilized GitHub Copilot to accelerate the SDLC, increasing efficiency in component development and unit test creation.
- **Security & Vulnerability Mitigation:** Remediated vulnerabilities detected by GitHub Advanced Security. Proactively managed third-party dependencies and performed code-level fixes to ensure compliance with enterprise information assurance standards.

SELECTED PROJECTS

Sinkhole - Procedural Dungeon Crawler
Jan 2026 - Present

- **Infinite World Architecture:** Engineered a chunk-based streaming system in Godot 4 (C#/.NET) that generates world data at runtime using seeded random algorithms.
- **Procedural Generation:** Developed custom layout logic to construct non-linear, endlessly descending floors, balancing randomness with navigable structural constraints.
- **Performance Optimization:** Decoupled generation logic from the main render thread to prevent frame-rate stutters during high-intensity procedural generation cycles.

World State AI - Resource Optimization Simulation
Aug 2025 - Oct 2025

- **Expected Utility Algorithm:** Engineered an incremental algorithm to evaluate and rank action schedules based on calculated expected utility and reward functions.
- **Decision Logic:** Implemented a parser to translate complex state transitions into executable schedules, identifying optimal resource trades to maximize global state stability.
- **Scalable Modeling:** Developed weights for diverse resource types, enabling the simulation of complex international trade and transformation resource loops.

Probabilistic Sensor Verification
Oct 2025 - Nov 2025

- **Formal Verification & Modeling:** Developed a framework to evaluate sensor reliability using PRISM model checker, modeling gas/thermal sensors as DTMC to simulate failure states.
- **Safety & Risk Assessment:** Authored PCTL safety properties to analyze system reachability and hazard thresholds, identifying theoretical failure points and safety boundaries.
- **Statistical Analysis:** Conducted cross-validation of formal results against Monte Carlo simulations using StormPy, evaluating variance between mathematical methods and empirical simulation.

Kubernetes Cluster for Distributed Systems
Oct 2024 - Nov 2024

- **Cloud Orchestration:** Architected a K8s cluster across four VMs on Chameleon Cloud. Resolved complex node-to-node networking and resource contention challenges.
- **Distributed Systems:** Integrated Apache Kafka for high-throughput messaging, optimizing resource provisioning to scale throughput by 10x (up to 5,000 messages per node).
- **Automation:** Authored custom Ansible playbooks and designed Dockerfiles to ensure environment parity and enforced deployment immutability via CI/CD.

Model-Integrated Computing - Turtle Graphics Studio
Oct 2024 - Dec 2024

- **Software Design:** Developed a visual design studio using WebGME, defining the DSML syntax and utilizing metamodeling to establish formal structural constraints.
- **Code Generation:** Engineered a model-to-code transformation pipeline to translate visual design elements into executable code, enabling rapid prototyping.
- **Formal QA:** Built custom plugins, including a Loop Detector, to perform formal automated verification, preventing infinite loops and ensuring strict constraint adherence.

PROFESSIONAL PORTFOLIO - JONATHANPOTEET.COM

- **Creative Engineering:** Combines formal software engineering with procedural generation and 3D design (Godot, Three.js, Blender) to build interactive, logic-driven environments.
- **Design-to-Code Mastery:** Features pixel-perfect implementations of complex UI mocks, showcasing architectural rigor in frontend development.

EDUCATION

Vanderbilt University — December 2025
Masters of Science in Computer Science - GPA: 3.97

Florida State University — December 2020
Bachelors of Science in Information Technology - Minor in Business - GPA: 3.80

CERTIFICATIONS

CompTIA Security+ (SY0-701) — 2026
Global Citizenship Certificate - Florida State University — 2020

SKILLS

Frontend/Web: Angular, Solid.js, React, Next.js, TypeScript, JavaScript, HTML, CSS/SCSS
Backend/Runtime: Node.js, Spring Boot, Kafka, Java, Python, C, C#, SQL
Databases: MongoDB (NoSQL), MySQL, Postgres (SQL)
DevOps/Tools: Kubernetes, Docker, Podman, Ansible, Azure DevOps, CI/CD, Jupyter, NPM, Jira
Networking & Creative: Linux, WSL, TCP/IP, Godot, Three.js, Procedural Algorithms

CAMPUS & COMMUNITY INVOLVEMENT

Study Abroad: FSU International Programs - London, England — Summer 2019
Air Force ROTC (AFROTC) - Detachment 145 — 2017 - 2018
Florida American Legion Boys State - Delegate — 2016
Eagle Scout, Boy Scouts of America — 2016